



**PAN3740 系列**

# 产品说明书

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上海磐启微电子有限公司

Sub-1G & 2.4G 多模无线 SOC 芯片



## PAN3740 Series Sub-1G & 2.4G Multi-mode Wireless SOC Chip

### Overview

The PAN3740 series is a SOC chip that integrates Sub-1G and 2.4G dual-band transceiver.

The Sub-1G band is a low-power, long-distance wireless transceiver chip that uses ChirpLoTTM modulation and demodulation technology. It supports half-duplex wireless communication and has operating frequency bands of 137~187MHz, 205~281MHz, 408~550MHz and 816~1110MHz. It has the characteristics of high anti-interference, high sensitivity, low power consumption and ultra-long transmission distance. It has a maximum sensitivity of -143dBm and a maximum output power of 20dBm, which produces an industry-leading link budget, making it the best choice for long-distance transmission and applications with extremely high reliability requirements. Compared with conventional modulation technology, PAN3740 also has significant advantages in blocking and adjacent channel selection, which can further improve communication reliability. At the same time, it also provides greater flexibility. Users can adjust the spread spectrum modulation bandwidth, spreading factor and error correction rate by themselves, effectively improving the compromise between distance, anti-interference ability and power consumption of chips using conventional modulation technology.

The 2.4G band supports BLE5.3 and 2.4GHz dual-mode operation, and operates in the 2.400 ~ 2.483GHz world-wide ISM band. At the same time, the PAN3740 series has built-in 512KB Flash program memory and 48KB SRAM memory. In addition, it is also equipped with a wealth of peripherals, including up to 19 GPIOs, 8 PWMs, 1 32-bit timer, 2 24-bit timers, 1 I2C, 2 UARTs, 2 SPIs, 7 external channels of ADC, WDT, WWDT, 32K RC automatic calibration, etc.

### Key Features

#### Sub-1G band

Working frequency band:

137~187MHz , 205~281MHz , 408~550MHz ,  
816~1110MHz

Modulation mode: ChirpLoTTM

Transmit output power: -30dBm~20dBm

Maximum link budget: 163dB

Sensitivity as low as: -143dBm@62.5kHz

Operating

current Sleep current:

260nA Receive current: 4.1mA@DCDC

mode Transmit

current:

96mA@20dBm

13mA@-2dBm Supported bandwidth

62.5kHz, 125kHz, 250kHz, 500kHz

Support SF factor (5~12), support automatic identification of

spreading factor Support code rate: 4/5,

4/6, 4/7, 4/8 Support

CAD function Support low rate mode:

0.08~59.9kbps Support 4-wire SPI / 3-wire SPI / I2C configuration interface,

GPIOs

#### 2.4G band

##### RF

operating frequency band: 2.400 ~

2.483GHz

Supported modes Various modes of BLE5.3: 1Mbps, 2Mbps,  
500kbps,

125kbps 2.4G private protocol, supports 1Mbps, 2Mbps,

500kbps, 250kbps, 125kbps, supports hardware

ACK Transmit power: up to 9dBm

Receiving sensitivity:

-99dBm@125kbps

-99dBm@500kbps

-96dBm@1Mbps

-93dBm@2Mbps RSSI

Resolution:

0.25dB Accuracy: ±2dB

Range: -90dBm ~

-15dBm Support single antenna

Safety regulations:

BQB/ETSI/FCC MCU 32-bit

MCU,

maximum frequency 48MHz Memory Flash:

built-in

512KB, support deep sleep mode SRAM: 48KB

eFuse: 128B

I-cache: 4KB Low

power

reception mode: 2.5mA@1Mbps (DCDC)

Transmission mode: 5.06mA@0dBm (DCDC)

Standby mode: 0.28uA

Standby mode (SRAM retention): 1.88uA (support RCL / XTL

and GPIO wake-up) Deep

sleep mode: 3.37uA (All Logic Retention /

GPIO / XTL / RCL can be woken

up) Clock

32MHz RC 32MHz

XTAL 32kHz RC

32.768kHz XTAL

DPLL (48MHz)

Peripherals up to 21

GPIOs

(IO has two supply voltages)



PAN3740 Series Sub-1G & 2.4G Multi-mode Wireless SOC Chip



8-channel PWM 1  
32-bit timer, 2 24-bit timers 1 I2C 2 UARTs 2 SPIs 2  
DMAs 12-  
channel ADC (7 ext,  
VBG\_1P2,  
VBG\_VT,  
  
1/4VDD, VBG\_0P6, Temp2y Support  
WDT/WWDT Support IO/BOD/  
POR/LVR/System reset FMC (support IAP, support boot  
loader with address 0x0)  
Program)  
Clock measurement, clock  
calibration Flash data encryption

yTemperature sensor  
support Temperature sensor  
detection range: -40y ~ 85y Package  
QFN48  
Working  
conditions  
Working temperature: -40y ~ 85y  
Working voltage: 2V ~ 3.6V (DCDC)

Typical Applications  
  
y Security, fire protection, medical treatment, metering, location, sensing, etc.

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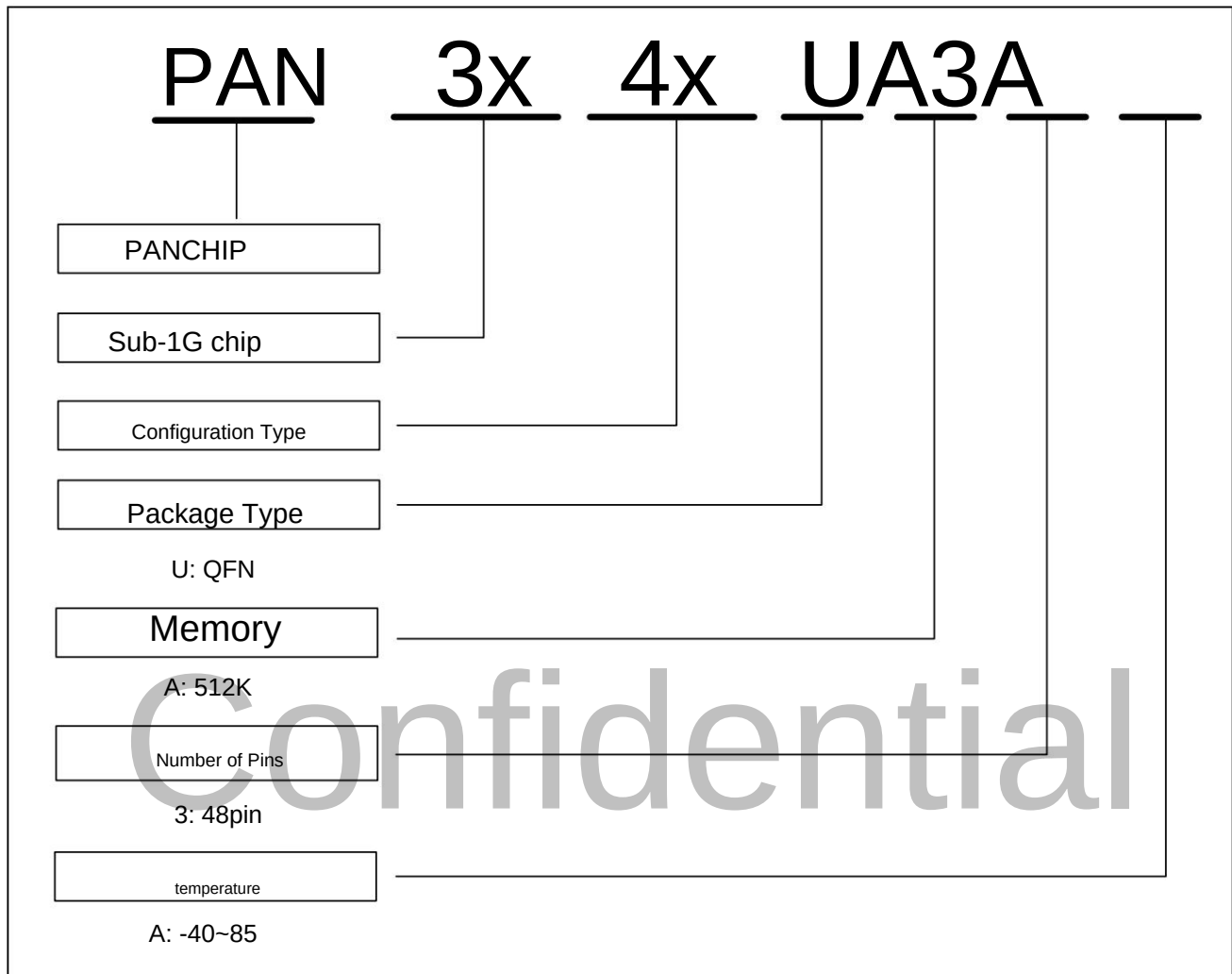
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## 1 Naming Rules





2Ordering Information

| Product Model | Chip Type  | Package Pin Number | IO FLASH RAM | Temperature | Packaging  |             |
|---------------|------------|--------------------|--------------|-------------|------------|-------------|
| PAN3740UA3A   | Sub-1G QFN | 48                 |              | 512K        | 48K -40~85 | Tape & Reel |

Before ordering, please consult sales for the latest mass production information.

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### 3 System structure diagram

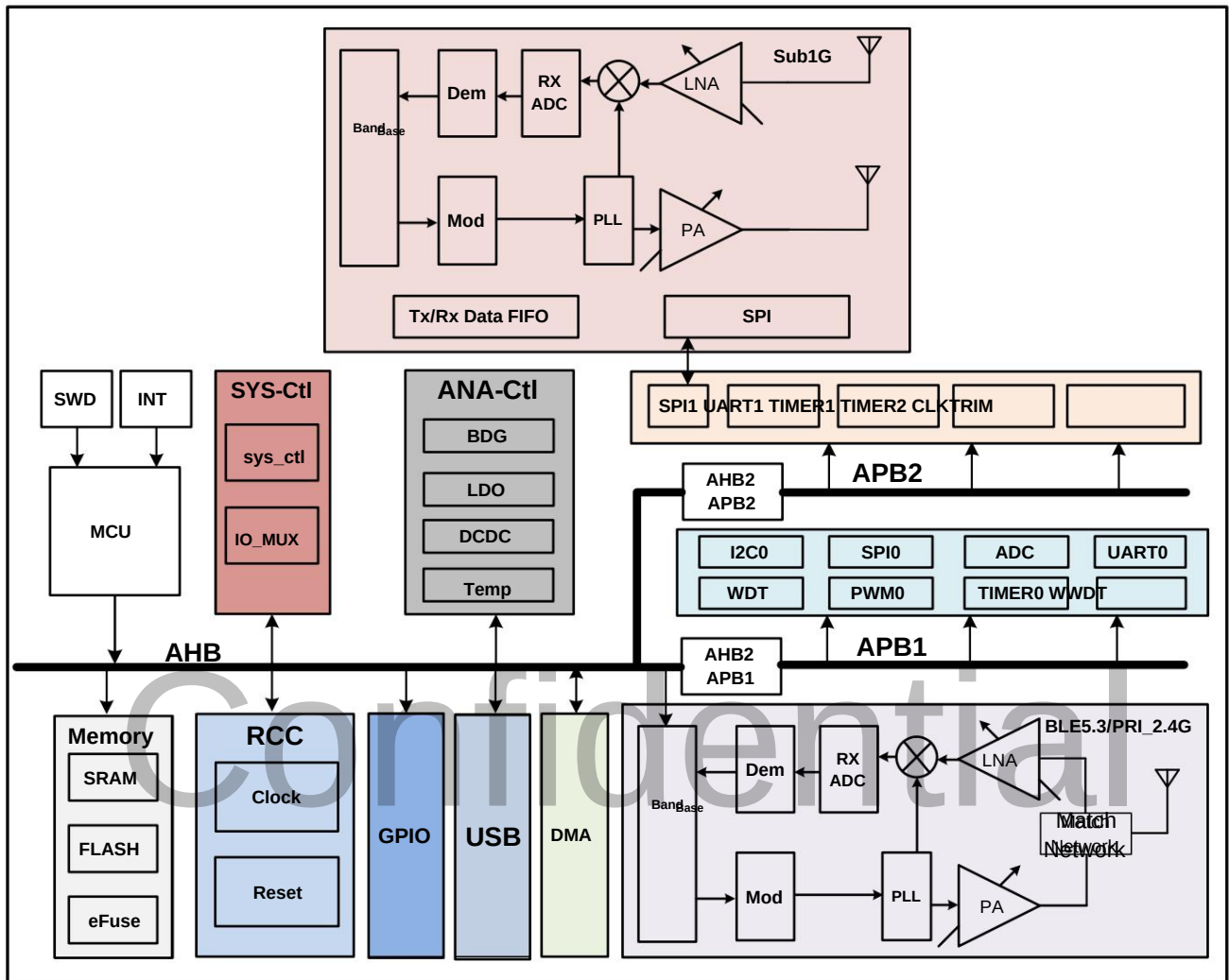


Figure 3-1 System structure diagram





4 Pin Definition and Description

4.1 Pin Diagram

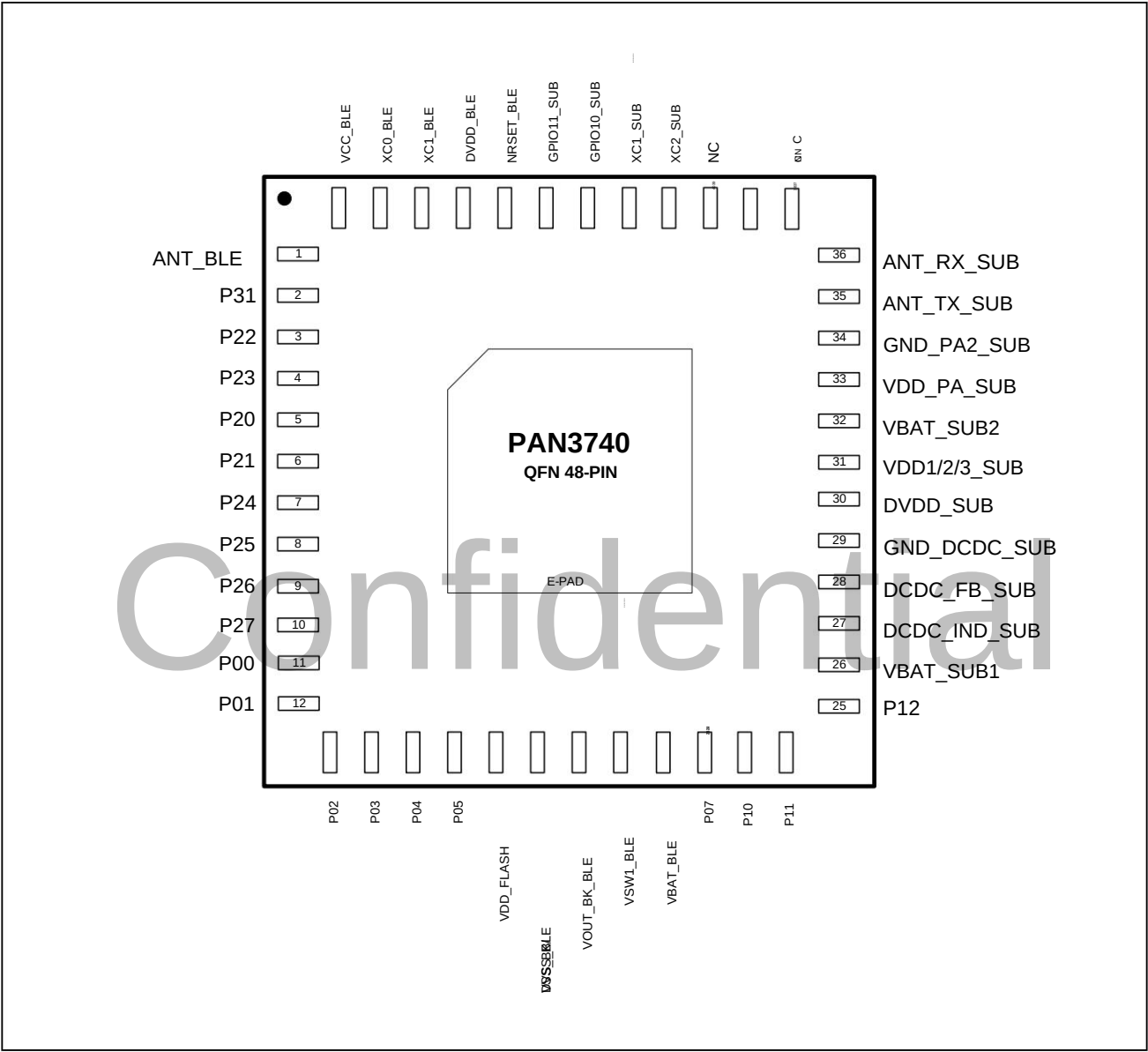


Figure 4-1 QFN48 pin diagram



## 4.2 Pin Description

Table 4-1 Pin Description

| Pin Number | Pin Name       | Pin Type         | Description                                                 |
|------------|----------------|------------------|-------------------------------------------------------------|
| 1          | ANT_BLE        | AI/AO RF antenna | RF antenna pin, an external antenna is required when in use |
| 2          | P31            | I/O              | General purpose digital input and output pins               |
|            | UART1_TX       | O                | UART1 TX pin                                                |
|            | PWM_CH7        | O                | PWM channel 7 output pin                                    |
|            | SPI1_MOSI      | I/O              | SPI1 MOSI pin                                               |
| 3          | P22            | I/O              | General purpose digital input and output pins               |
|            | SPI1_CLK       | I/O              | SPI1 clock pin                                              |
|            | PWM_CH0        | O                | PWM channel 0 output pin                                    |
|            | SPI0_MOSI      | I/O              | SPI0 MOSI pin                                               |
|            | ADC_CH2        | AI               | ADC Channel 2 Pin                                           |
| 4          | P23            | I/O              | General purpose digital input and output pins               |
|            | SPI1_CS        | I/O              | SPI1 CS pin                                                 |
|            | PWM_CH1        | O                | PWM channel 1 output pin                                    |
|            | DPLL_DIV8      | O                | Dpll_div8 output pin                                        |
|            | I2C0_SCL       | I/O              | I2C0 SCL pin                                                |
|            | ADC_CH1        | AI               | ADC Channel 1 Pin                                           |
| 5          | P20            | I/O              | General purpose digital input and output pins               |
|            | SPI1_MISO      | I/O              | SPI1 MISO pin                                               |
|            | PWM_CH6        | O                | PWM channel 6 output pin                                    |
| 6          | P21            | I/O              | General purpose digital input and output pins               |
|            | XTL0           | AI               | External 32.768kHz clock source output                      |
|            | SPI1_MOSI      | I/O              | SPI1 MOSI pin                                               |
|            | PWM_CH7        | O                | PWM channel 7 output pin                                    |
| 7          | P24            | I/O              | General purpose digital input and output pins               |
|            | UART1_RX       | I                | UART1 RX pin                                                |
|            | SPI1_MISO      | I/O              | SPI1 MISO pin                                               |
|            | PWM_CH2        | O                | PWM channel 2 output pin                                    |
|            | TIMER1_CNT_OUT | O                | TIMER1 output pin                                           |
|            | ADC_CH6        | AI               | ADC Channel 6 Pin                                           |
| 8          | P25            | I/O              | General purpose digital input and output pins               |



| Pin Number | Pin Name  | Pin Type | Description                                   |
|------------|-----------|----------|-----------------------------------------------|
|            | UART1_TX  | O        | UART1 TX pin                                  |
|            | SPI1_MOSI | I/O      | SPI1 MOSI pin                                 |
|            | PWM_CH3   | O        | PWM channel 3 output pin                      |
|            | I2C0_SDA  | I/O      | I2C0 SDA pin                                  |
|            | ADC_CH7   | AI       | ADC Channel 7 Pin                             |
| 9          | P26       | I/O      | General purpose digital input and output pins |
|            | UART1_RTS | O        | UART1 RTS pin                                 |
|            | PWM_CH4   | O        | PWM channel 4 output pin                      |
|            | SPI1_CS   | I/O      | SPI1 CS pin                                   |
| 10         | P27       | I/O      | General purpose digital input and output pins |
|            | UART1_CTS | I        | UART1 CTS pin                                 |
|            | PWM_CH5   | O        | PWM channel 5 output pin                      |
|            | SPI1_CLK  | I/O      | SPI1 clock pin                                |
|            | XTH       | O        | Xth output pin                                |
| 11         | P00       | I/O      | General purpose digital input and output pins |
|            | SWD_CLK   | I        | SWD clock input pin                           |
|            | UART1_RX  | I        | UART1 RX pin                                  |
|            | I2C0_SCL  | I/O      | I2C0 SCL pin                                  |
|            | SPI0_CLK  | I/O      | SPI0 clock pin                                |
| 12         | P01       | I/O      | General purpose digital input and output pins |
|            | SWD_DAT   | I/O      | SWD data input and output pin                 |
|            | UART1_TX  | O        | UART1 TX pin                                  |
|            | I2C0_SDA  | I/O      | I2C0 SDA pin                                  |
|            | SPI0_CS   | I/O      | SPI0 CS pin                                   |
| 13         | P02       | I/O      | General purpose digital input and output pins |
|            | UART0_CTS | I        | UART0 CTS pin                                 |
|            | SPI1_MISO | I/O      | SPI1 MISO pin                                 |
|            | PWM_CH3   | O        | PWM channel 3 output pin                      |
|            | UART1_CTS | I        | UART1 CTS pin                                 |
| 14         | P03       | I/O      | General purpose digital input and output pins |
|            | UART0_CTS | I        | UART0 CTS pin                                 |
|            | SPI0_CS   | I/O      | SPI0 CS pin                                   |
|            | PWM_CH2   | O        | PWM channel 2 output pin                      |



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| Pin Number | Pin Name       | Pin Type | Description                                                                                         |
|------------|----------------|----------|-----------------------------------------------------------------------------------------------------|
|            | TIMER0_CNT_OUT | O        | TIMER0 output pin                                                                                   |
| 15         | P04            | I/O      | General purpose digital input and output pins                                                       |
|            | UART0_RTS      | O        | UART0 RTS pin                                                                                       |
|            | SPI0_CLK       | I/O      | SPI0 clock pin                                                                                      |
|            | PWM_CH3        | O        | PWM channel 3 output pin                                                                            |
|            | TM0_EXT        | I        | TIMER0 external input pin                                                                           |
|            | ADC_CH0        | AI       | ADC Channel 0 Pin                                                                                   |
| 16         | P05            | I/O      | General purpose digital input and output pins                                                       |
|            | UART0_TX       | O        | UART0 TX pin                                                                                        |
|            | SPI0_MISO      | I/O      | SPI0 MISO pin                                                                                       |
|            | PWM_CH4        | O        | PWM channel 4 output pin                                                                            |
| 17         | VDD_FLASH      | P        | FLASH Power Supply                                                                                  |
| 18         | DVSS_BLE       | P        | Digitally                                                                                           |
|            | VSS_BK_BLE     | P        | Common ground terminal of DCDC power supply, independent power ground                               |
| 19         | VOUT_BK_BLE    | P        | DCDC voltage output                                                                                 |
| 20         | VSW1_BLE       | P        | DCDC internal power switch (switching frequency about 650kHz), required when used External inductor |
| VBAT_BLE   |                | P        | Power input pin of the chip                                                                         |
| P07        | P07            | I/O      | General purpose digital input and output pins                                                       |
|            | UART1_RX       | I        | UART1 RX pin                                                                                        |
|            | I2C0_SCL       | I/O      | I2C0 clock pin                                                                                      |
|            | SPI0_MOSI      | I/O      | SPI0 MOSI pin                                                                                       |
|            | PWM_CH0        | O        | PWM channel 0 output pin                                                                            |
|            | ADC_CH5        | AI       | ADC channel 5 pin                                                                                   |
| P10        | P10            | I/O      | General purpose digital input and output pins                                                       |
|            | UART1_TX       | O        | UART1 TX pin                                                                                        |
|            | I2C0_SDA       | I/O      | I2C0 SDA pin                                                                                        |
|            | SPI0_MISO      | I/O      | SPI0 MISO pin                                                                                       |
|            | PWM_CH5        | O        | PWM channel 5 output pin                                                                            |
| P11        | P11            | I/O      | General purpose digital input and output pins                                                       |
|            | UART1_RTS      | O        | UART1 RTS pin                                                                                       |
|            | SPI0_MOSI      | I/O      | SPI0 MOSI pin                                                                                       |
|            | PWM_CH7        | O        | PWM channel 7 output pin                                                                            |



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| Pin Number | Pin Name       | Pin Type | Description                                                                                  |
|------------|----------------|----------|----------------------------------------------------------------------------------------------|
|            | CLK_32K        | O        | Clk_32k output pin                                                                           |
|            | UART0_TX       | O        | UART0 TX pin                                                                                 |
| 25         | P12            | I/O      | General purpose digital input and output pins                                                |
|            | UART0_RX       | I        | UART0 RX pin                                                                                 |
|            | TIMER0_CNT_OUT | O        | TIMER0 output pin                                                                            |
|            | UART1_TX       | O        | UART1 TX pin                                                                                 |
|            | SPI0_MISO      | I/O      | SPI0 MISO pin                                                                                |
|            | PWM_CH4        | O        | PWM channel 4 output pin                                                                     |
|            | ADC_CH3        | AI       | ADC Channel 3 Pin                                                                            |
| 26         | VBAT_SUB1      | P        | Digital GPIO power supply / DCDC power supply, connected to the main power supply            |
| 27         | DCDC_IND_SUB   | AO       | Internal DCDC output, connect external series inductor                                       |
| 28         | DCDC_FB_SUB    | AI       | Internal DCDC feedback input, DCDC mode connected to VDD1/2/3                                |
| 29         | GND_DCDC_SUB   | G        | Analog Ground                                                                                |
| 30         | DVDD_SUB       | P        | Digital power LDO output                                                                     |
| 31         | VDD1/2/3_SUB   | P        | Analog power supply, DCDC mode connects to VFB, non-DCDC mode connects to total power supply |
| 32         | VBAT_SUB2      | P        | Analog power supply, connected to the main power supply                                      |
| 33         | VDD_PA_SUB     | P        | PA power LDO output                                                                          |
| 34         | GND_PA2_SUB    | P        | PA analog ground                                                                             |
| 35         | ANT_TX_SUB     | AO       | Transmitter PA Output                                                                        |
| 36         | ANT_RX_SUB     | AI       | Receiver LNA Input                                                                           |
| 37         | NC             |          |                                                                                              |
| 38         | NC             |          |                                                                                              |
| 39         | NC             |          |                                                                                              |
| 40         | XC2_SUB        | AI       | Crystal input                                                                                |
| 41         | XC1_SUB        | AO       | Crystal output                                                                               |
| 42         | GPIO10_SUB     | I/O      | Digital IO                                                                                   |
|            |                | O        | External PA control signal                                                                   |
| 43         | GPIO11_SUB     | I/O      | Digital IO                                                                                   |
|            |                | O        | Channel status indicator signal                                                              |
| 44         | NRSET_BLE      | I        | Reset pin                                                                                    |
| 45         | DVDD_BLE       | P        | HLDO output pin, typical value 1.2V                                                          |
| 46         | XC1_BLE        | AO       | External 32MHz clock source output                                                           |



| Pin Number | Pin Name | Pin Type | Description                             |
|------------|----------|----------|-----------------------------------------|
| 47         | XC0_BLE  | AI       | External 32MHz clock source input       |
| 48         | VCC_BLE  | P        | Chip power supply                       |
| 49         | E-PAD    | P        | Chip bottom pad, common ground terminal |

4.3 Internal Connections

Table 4-2 Internal connection pin description

| Pin Status | RF          | MCU |
|------------|-------------|-----|
| IS         | PAD_MOSI_3V | P16 |
| IS         | PAD_MISO_3V | P30 |
| IS         | PAD_CSN_3V  | P14 |
| IS         | PAD_SCK_3V  | P15 |
| IS         | PAD_IRQ_3V  | P17 |
| IS         | NRSET       | P13 |

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5 Electrical characteristics

Maximum and minimum values

The notes below each table indicate that the data are obtained through comprehensive evaluation, design simulation and/or process characterization.

The test will be carried out on the production line; based on the comprehensive evaluation, the minimum and maximum values are obtained after the sample test.

The mean value is obtained by adding or subtracting three times the standard distribution (mean ± 3σ).

5.1 Sub-1G frequency band

Note:

The VDD and other parameters described in this section correspond to the relevant power input pins such as "xxx\_SUB" in the pin diagram.

All parameters in this section are based on TA=25℃, power supply voltage 3.3V test, unless otherwise specified.

5.1.1 Absolute Maximum Ratings

Table 5-1 Absolute maximum ratings

| Symbolic parameters |                             | condition         | Min  | Typ | Max | Unit |
|---------------------|-----------------------------|-------------------|------|-----|-----|------|
| VDD                 | VDD1/VDD2/VDD3/VBAT/VBAT_IO | Frequency: 490MHz | -0.3 | 3.3 | 3.6 | V    |
| VI                  | Input voltage               |                   | -0.3 | -   | VDD | V    |
| VO                  | Output voltage              |                   | VSS  | -   | VDD | V    |
| TOP                 | Operating temperature       |                   | -40  | -   | 85  | ℃    |
| TSTG storage        | temperature                 |                   | -55  | -   | 125 | ℃    |

Notice:

- Exceeding one or more of the limiting values may cause permanent damage to the chip.
- Electrostatically sensitive device, comply with protection regulations when handling.



## 5.1.2 DC characteristics

Test conditions:

Frequency: 490MHz

Table 5-2 Voltage and current (frequency 490MHz)

| symbol     | parameter                   | Min     | Typ | Max | Unit | Test Conditions                    |
|------------|-----------------------------|---------|-----|-----|------|------------------------------------|
| VDD        | power supply                | 1.8     |     | 3.3 | V    | TA=25℃, LDO mode                   |
|            |                             | 2       |     | 3.3 | V    | TA=25℃, DCDC mode                  |
| VSS        | land                        | -       |     | 0   | V    | -                                  |
| IDeepSleep | deep sleep current          | -       |     | 260 | nA   | -                                  |
| ISleep     | Sleep current               | -       |     | 620 | nA   | -                                  |
| ITX,20dBm  | TX mode operating current - |         |     | 96  | mA   | 20dBm output power                 |
| ITX,18dBm  | TX mode operating current - |         |     | 77  | mA   | 18dBm output power                 |
| ITX,-2dBm  | TX mode operating current - |         |     | 13  | mA   | -2dBm output power                 |
| IRX,LDO    | RX mode operating current - |         |     | 7.8 | mA   | In LDO mode, the maximum LNA Gain  |
| IRX,DCDC   | RX mode operating current - |         |     | 4.1 | mA   | In DCDC mode, the maximum LNA Gain |
| VOH        | Output high level voltage   | VDD-0.3 |     |     | V    | -                                  |
| VOL        | Output low level voltage    | VSS     |     |     | V    | -                                  |
| VIH        | Input high level voltage    | 0.8*VDD |     |     | V    | -                                  |
| VIL        | Input low level voltage -   |         |     |     | V    | -                                  |
| SPI_rate   | SPI Rate                    | -       |     | 10  | Mbps | -                                  |

Test conditions:

Frequency: 915MHz

Table 5-3 Voltage and current (frequency 915MHz)

| symbol     | parameter                   | Min | Typ | Max | Unit | Test Conditions    |
|------------|-----------------------------|-----|-----|-----|------|--------------------|
| VDD        | power supply                | 1.8 |     | 3.3 | V    | TA=25℃, LDO mode   |
|            |                             | 2   |     | 3.3 | V    | TA=25℃, DCDC mode  |
| VSS        | land                        | -   |     | 0   | V    | -                  |
| IDeepSleep | deep sleep current          | -   |     | 260 | nA   | -                  |
| ISleep     | Sleep current               | -   |     | 620 | nA   | -                  |
| ITX,20dBm  | TX mode operating current - |     |     | 100 | mA   | 20dBm output power |




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|              |                             |           |     |           |        |                                    |
|--------------|-----------------------------|-----------|-----|-----------|--------|------------------------------------|
| ITX,18dBm TX | mode operating current -    |           | 81  | -         | mA     | 18dBm output power                 |
| ITX,-2dBm    | TX mode operating current - |           | 15  | -         | mA     | -2dBm output power                 |
| IRX,LDO      | RX mode operating current - |           | 7.8 | -         | mA     | In LDO mode, the maximum LNA Gain  |
| IRX,DCDC     | RX mode operating current - |           | 4.1 | -         | mA     | In DCDC mode, the maximum LNA Gain |
| VOH          | Output high level voltage   | VDD-0.3   | -   | VDD       | V      | -                                  |
| VOL          | Output low level voltage    | VSS       | -   | VSS+0.3 V |        | -                                  |
| VIH          | Input high level voltage    | 0.8*VDD - |     | -         | V      | -                                  |
| VIL          | Input low level voltage -   |           | -   | 0.2*VDD V |        | -                                  |
| SPI_rate     | SPI Rate                    | -         | -   | 10        | Mbps - |                                    |

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## 5.1.3 RF Performance

Test conditions:

Frequency: 490MHz

γ Error correction code = 4/8

γ Packet error rate γ 5%

γ Payload length = 10Bytes

Table 5-4 General RF characteristics (frequency 490MHz)

| symbol              | describe                                                                                                   | Conditions | Min  | Typ  | Max      | Unit |
|---------------------|------------------------------------------------------------------------------------------------------------|------------|------|------|----------|------|
| Universal frequency |                                                                                                            |            |      |      |          |      |
| Fop                 | Operating frequency                                                                                        | -          | 137  | -    | 187      | MHz  |
|                     |                                                                                                            | -          | 205  | -    | 281      | MHz  |
|                     |                                                                                                            | -          | 408  | -    | 550      | MHz  |
|                     |                                                                                                            | -          | 816  | -    | 1110     | MHz  |
| Fxtal               | Crystal frequency                                                                                          | -          | -    | 32   | -        | MHz  |
| RS                  | Crystal Series Resistors                                                                                   | -          | -    | 30   | 50       | γ    |
| CFOOT               | crystal external capacitance                                                                               | -          | 8    | 15   | infinity | pF   |
| CLOAD               | Crystal Load Capacitance                                                                                   | -          | 6    | 10   | 12       | pF   |
| FTOL                | Initial Frequency                                                                                          | -          | -    | ±10  | -        | ppm  |
| BR                  | Tolerance Bit Rate                                                                                         | -          | 0.08 | -    | 59.9     | kbps |
| Transmitter         |                                                                                                            |            |      |      |          |      |
| PLPWAN              | output power                                                                                               | -          | -30  | -    | 20       | dBm  |
| Receiver            |                                                                                                            |            |      |      |          |      |
| RF_62.5<br>(LDO)    | RF sensitivity, long range mode, maximum LNA gain,<br>Uses separate RX/TX channels with 62.5 kHz bandwidth | SF = 7     | -    | -129 | -        | dBm  |
|                     |                                                                                                            | SF = 10    | -    | -137 | -        |      |
|                     |                                                                                                            | SF = 12    | -    | -143 | -        |      |
| RF_125<br>(LDO)     | RF sensitivity, long range mode, maximum LNA gain,<br>Uses separate RX/TX channels 125 kHz bandwidth       | SF = 7     | -    | -126 | -        | dBm  |
|                     |                                                                                                            | SF = 10    | -    | -134 | -        |      |
|                     |                                                                                                            | SF = 12    | -    | -140 | -        |      |
| RF_250<br>(LDO)     | RF sensitivity, long range mode, maximum LNA gain,<br>Uses separate RX/TX channels 250 kHz bandwidth       | SF = 7     | -    | -124 | -        | dBm  |
|                     |                                                                                                            | SF = 10    | -    | -132 | -        |      |
|                     |                                                                                                            | SF = 12    | -    | -137 | -        |      |
| RF_500<br>(LDO)     | RF sensitivity, long range mode, maximum LNA gain,<br>Uses separate RX/TX channels 500 kHz bandwidth       | SF = 7     | -    | -121 | -        | dBm  |
|                     |                                                                                                            | SF = 10    | -    | -129 | -        |      |
|                     |                                                                                                            | SF = 12    | -    | -134 | -        |      |
| RF_62.5<br>(DCDC)   | RF sensitivity, long range mode, maximum LNA gain,<br>Uses separate RX/TX channels with 62.5 kHz bandwidth | SF = 7     | -    | -125 | -        | dBm  |
|                     |                                                                                                            | SF = 10    | -    | -133 | -        |      |
|                     |                                                                                                            | SF = 12    | -    | -139 | -        |      |
| RF_125<br>(DCDC)    | RF sensitivity, long range mode, maximum LNA gain,<br>Uses separate RX/TX channels 125 kHz bandwidth       | SF = 7     | -    | -122 | -        | dBm  |
|                     |                                                                                                            | SF = 10    | -    | -130 | -        |      |
|                     |                                                                                                            | SF = 12    | -    | -137 | -        |      |





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|                  |                                                                                                      |                              |             |                      |             |     |
|------------------|------------------------------------------------------------------------------------------------------|------------------------------|-------------|----------------------|-------------|-----|
| RF_250<br>(DCDC) | RF sensitivity, long range mode, maximum LNA gain,<br>Uses separate RX/TX channels 250 kHz bandwidth | SF = 7<br>SF = 10<br>SF = 12 | -<br>-<br>- | -120<br>-128<br>-134 | -<br>-<br>- | dBm |
| RF_500<br>(DCDC) | RF sensitivity, long range mode, maximum LNA gain,<br>Uses separate RX/TX channels 500 kHz bandwidth | SF = 7<br>SF = 10<br>SF = 12 | -<br>-<br>- | -117<br>-125<br>-130 | -<br>-<br>- | dBm |

Test conditions:

Frequency: 915MHz

γ Error correction code = 4/8

γ Packet error rate γ 5%

γ Payload length = 10Bytes

Table 5-5 General RF characteristics (frequency 915MHz)

| symbol              | describe                                                                                                   | Conditions                   | Min         | Typ                  | Max         | Unit |
|---------------------|------------------------------------------------------------------------------------------------------------|------------------------------|-------------|----------------------|-------------|------|
| Universal frequency |                                                                                                            |                              |             |                      |             |      |
| Fop                 | Operating frequency                                                                                        | -                            | 137         | -                    | 187         | MHz  |
|                     |                                                                                                            | -                            | 205         | -                    | 281         | MHz  |
|                     |                                                                                                            | -                            | 408         | -                    | 550         | MHz  |
|                     |                                                                                                            | -                            | 816         | -                    | 1110        | MHz  |
| Fxtal               | Crystal frequency                                                                                          | -                            | -           | 32                   | -           | MHz  |
| RS                  | Crystal series resistance                                                                                  | -                            | -           | 30                   | 50          | γ    |
| CFOOT               | crystal external capacitance                                                                               | -                            | 8           | 15                   | -           | pF   |
| CLOAD               | Crystal load capacitance initial                                                                           | -                            | 6           | 10                   | 12          | pF   |
| ETOL                | frequency tolerance                                                                                        | -                            | -           | ±10                  | -           | ppm  |
| BR                  | bit rate                                                                                                   | -                            | 0.08        | -                    | 59.9        | kbps |
| Transmitter         |                                                                                                            |                              |             |                      |             |      |
| PLPWAN              | output power                                                                                               | -                            | -30         | -                    | 20          | dBm  |
| Receiver            |                                                                                                            |                              |             |                      |             |      |
| RF_62.5<br>(LDO)    | RF sensitivity, long range mode, maximum LNA gain,<br>Uses separate RX/TX channels with 62.5 kHz bandwidth | SF = 7<br>SF = 10<br>SF = 12 | -<br>-<br>- | -128<br>-136<br>-141 | -<br>-<br>- | dBm  |
| RF_125<br>(LDO)     | RF sensitivity, long range mode, maximum LNA gain,<br>Uses separate RX/TX channels 125 kHz bandwidth       | SF = 7<br>SF = 10<br>SF = 12 | -<br>-<br>- | -125<br>-132<br>-138 | -<br>-<br>- | dBm  |
| RF_250<br>(LDO)     | RF sensitivity, long range mode, maximum LNA gain,<br>Uses separate RX/TX channels 250 kHz bandwidth       | SF = 7<br>SF = 10<br>SF = 12 | -<br>-<br>- | -121<br>-129<br>-135 | -<br>-<br>- | dBm  |
| RF_500<br>(LDO)     | RF sensitivity, long range mode, maximum LNA gain,<br>Uses separate RX/TX channels 500 kHz bandwidth       | SF = 7<br>SF = 10<br>SF = 12 | -<br>-<br>- | -119<br>-127<br>-132 | -<br>-<br>- | dBm  |
| RF_62.5             | RF sensitivity, long range mode, maximum LNA gain, SF = 7                                                  | -                            | -           | -128                 | -           | dBm  |


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|                                                         |                                                                                                       |         |   |      |   |     |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------|---|------|---|-----|
| (DCDC) Using separate RX/TX channels 62.5 kHz bandwidth |                                                                                                       | SF = 10 | - | -136 | - |     |
|                                                         |                                                                                                       | SF = 12 | - | -140 | - |     |
|                                                         |                                                                                                       |         |   |      |   |     |
| RF_125<br>(DCDC)                                        | RF sensitivity, long range mode, maximum LNA gain,<br>using separate RX/TX channels 125 kHz bandwidth | SF = 7  | - | -125 | - | dBm |
|                                                         |                                                                                                       | SF = 10 | - | -132 | - |     |
|                                                         |                                                                                                       | SF = 12 | - | -137 | - |     |
| RF_250<br>(DCDC)                                        | RF sensitivity, long range mode, maximum LNA gain,<br>using separate RX/TX channels 250 kHz bandwidth | SF = 7  | - | -120 | - | dBm |
|                                                         |                                                                                                       | SF = 10 | - | -129 | - |     |
|                                                         |                                                                                                       | SF = 12 | - | -135 | - |     |
| RF_500<br>(DCDC)                                        | RF sensitivity, long range mode, maximum LNA gain,<br>using separate RX/TX channels 500 kHz bandwidth | SF = 7  | - | -118 | - | dBm |
|                                                         |                                                                                                       | SF = 10 | - | -126 | - |     |
|                                                         |                                                                                                       | SF = 12 | - | -130 | - |     |

# Confidential



## 5.2 2.4G frequency band

Note:

The VDD and other parameters described in this section correspond to the relevant power input pins such as "xxx\_BLE" in the pin diagram.

### 5.2.1 Absolute Maximum Ratings

Table 5-6 Absolute maximum ratings

| Symbol           | Description               | Conditions                             | Parameter |     |             | Unit |
|------------------|---------------------------|----------------------------------------|-----------|-----|-------------|------|
|                  |                           |                                        | Min       | Typ | Max         |      |
| VDD - VSS Supply | voltages                  | TA=25°C                                | -0.3      | -   | 3.6         | V    |
| VIN              | I/O pin voltage           | TA=25°C                                | VSS-0.3 - |     | VDD + 0.3 V |      |
| PVDD             | Extreme power consumption | VDD=3.3V, TA=25°C<br>DCDC power supply | -         | -   | TBD         | mW   |

### 5.2.2 Normal working conditions

Table 5-7 General working conditions

| Symbol           | Description          | Conditions          | Parameter |     |       | Unit |
|------------------|----------------------|---------------------|-----------|-----|-------|------|
|                  |                      |                     | Min       | Typ | Max   |      |
| VDD*             | Operating voltage    | TA=25°C             | 1.8       | -   | 3.6 V |      |
| VIPIO2*          | Operating voltage    | TA=25°C             | 1.8       | -   | 3.6 V |      |
| TST              | Storage temperature  | -                   | -65       | -   | 150   | °C   |
| TA               | Ambient temperature  | -                   | -40       | -   | 125   | °C   |
| TJ               | Junction temperature | QFN32 4x4x0.75-P0.4 | -40       | -   | 125   | °C   |
| R <sub>θJA</sub> | Thermal resistance   | QFN32 4x4x0.75-P0.4 | -         | TBD | -     | °C/W |

Note: DCDC-OFF

### 5.2.3 RF characteristics

Table 5-8 RF characteristics

| Symbol               | Description                         | Conditions | Parameter |      |          | Unit |
|----------------------|-------------------------------------|------------|-----------|------|----------|------|
|                      |                                     |            | Min       | Type | Max      |      |
| f                    | Operating frequency                 |            | 2400      | -    | 2483 MHz |      |
| PLLres               | PLL programming resolution          |            | 244       | 1M   | -        | Hz   |
| DR                   | Data rate                           |            | 0.125     | -    | 2        | Mbps |
| ƒ <sub>fBLE,2M</sub> | Frequency deviation@BLE 2Mbps       |            | 450       | 500  | 550      | kHz  |
| ƒ <sub>fBLE,1M</sub> | Frequency deviation @ BLE 1Mbps     |            | 225       | 250  | 275      | kHz  |
| ƒ <sub>f297,2M</sub> | Frequency deviation @ 297mode 2Mbps |            | 450       | 500  | 550      | kHz  |


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|                     |                                    |  |     |     |     |     |
|---------------------|------------------------------------|--|-----|-----|-----|-----|
| $\gamma f_{297,1M}$ | Frequency deviation@297mode 1Mbps  |  | 225 | 250 | 275 | kHz |
| $\gamma f_{N,2M}$   | Frequency deviation @ N-mode 2Mbps |  | -   | 320 | -   | kHz |
| $\gamma f_{N,1M}$   | Frequency deviation @ N-mode 1Mbps |  | -   | 170 | -   | kHz |
| $\gamma f_{BLE,2M}$ | Channel spacing @ BLE 2Mbps        |  | -   | 2   | -   | MHz |
| $\gamma f_{BLE,1M}$ | Channel spacing @ BLE 1Mbps        |  | -   | 2   | -   | MHz |
| $\gamma f_{297,2M}$ | Channel spacing@297mode 2Mbps      |  | -   | 2   | -   | MHz |
| $\gamma f_{297,1M}$ | Channel spacing@297mode 1Mbps      |  | -   | 1   | -   | MHz |
| $\gamma f_{N,2M}$   | Channel spacing @ N-mode 2Mbps     |  | -   | 2   | -   | MHz |
| $\gamma f_{N,1M}$   | Channel spacing @ N-mode 1Mbps     |  | -   | 1   | -   | MHz |

Table 5-9 TX characteristics

| Symbol   | Description                                | Conditions | Parameter |     |         | Unit |
|----------|--------------------------------------------|------------|-----------|-----|---------|------|
|          |                                            |            | Min       | Typ | Max     |      |
| PRFTX    | Output power                               |            | -         | -   | 9       | dBm  |
| PRFC     | RF power control range                     |            | -         | 40  | -       | dB   |
| PRFCR    | RF power accuracy                          |            | -         | -   | ±3      | dB   |
| PRF1M,1  | 1st Adjacent Channel Transmit Power @1Mbps |            | -         | -40 | -       | dBc  |
| PRF1M,2  | 2nd Adjacent Channel Transmit Power @1Mbps |            | -         | -56 | -       | dBc  |
| PRF1M, 3 | 3rd Adjacent Channel Transmit Power @1Mbps |            | -         | -60 | -       | dBc  |
| PRF2M,2  | 1st Adjacent Channel Transmit Power @2Mbps |            | -         | -37 | -       | dBc  |
| PRF2M,4  | 2nd Adjacent Channel Transmit Power @2Mbps |            | -         | -58 | -       | dBc  |
| PRF2M, 6 | 3rd Adjacent Channel Transmit Power @2Mbps |            | -         | -60 | -       | dBc  |
| PBW1M    | 20dB bandwidth @1Mbps                      |            | -         | 1.3 | -       | MHz  |
| PBW2M    | 20dB bandwidth @2Mbps                      |            | -         | 2.3 | -       | MHz  |
| PSP,1    | Spurious @1GHz                             |            | -         | -   | -73 dBm |      |
| PSP,2    | Spurious @1GHz                             |            | -         | -   | -49 dBm |      |

Table 5-10 RX characteristics

| Symbol         | Description                 | Conditions | Parameter |     |     | Unit |
|----------------|-----------------------------|------------|-----------|-----|-----|------|
|                |                             |            | Min       | Typ | Max |      |
| PRX,MIX        | Receive maximum input power |            | -         | 0   | -   | dBm  |
| PSENS,1M,BLE   | Sensitivity, 1Mbps BLE      |            | -         | -96 | -   | dBm  |
| PSENS,2M,BLE   | Sensitivity, 2Mbps BLE      |            | -         | -93 | -   | dBm  |
| PSENS,125K,BLE | Sensitivity, 125kbps BLE    |            | -         | -99 | -   | dBm  |
| PSENS,500K,BLE | Sensitivity, 500kbps BLE    |            | -         | -99 | -   | dBm  |



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|                       |                                                                   |                                                                          |   |     |   |     |
|-----------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------|---|-----|---|-----|
| PSENS,250K,B          | Sensitivity, 250kbps B mode                                       | Sensitivity, 1Mbps ideal transmitter, 37 bytes, BER = 0.1% is presented. | - | -99 | - | dBm |
| PSENS,1M,297          | Sensitivity, 1Mbps 297 mode                                       |                                                                          | - | -93 | - | dBm |
| PSENS,2M,297          | Sensitivity, 2Mbps 297 mode                                       |                                                                          | - | -91 | - | dBm |
| PSENS,250K,297        | Sensitivity, 250kbps 297 mode                                     |                                                                          | - | -99 | - | dBm |
| PSENS,1M,N            | Sensitivity, 1Mbps N-mode                                         |                                                                          | - | -93 | - | dBm |
| PSENS,2M,N            | Sensitivity, 2Mbps N-mode                                         |                                                                          | - | -91 | - | dBm |
| PSENS,250K,N          | Sensitivity, 250kbps N-mode                                       |                                                                          | - | -99 | - | dBm |
| C/I CO,1M,BLE         | Co-Channel interference@1Mbps                                     |                                                                          | - | 6   | - | dB  |
| C/I 1M,1M,BLE         | Adjacent (1 MHz) interference@1Mbps                               |                                                                          | - | -3  | - | dB  |
| C/I 2M,1M,BLE         | Adjacent (2 MHz) interference @1Mbps                              |                                                                          | - | -40 | - | dB  |
| C/I 3M,1M,BLE         | Adjacent (3 MHz) interference @1Mbps                              |                                                                          | - | -45 | - | dB  |
| C/I Image,1M,BLE      | Image frequency interference @1Mbps                               |                                                                          | - | -   | - | dB  |
| C/I Image±1M,1M,BLE   | Adjacent (±1MHz) interference to in-band image frequency @1Mbps   |                                                                          | - | -38 | - | dB  |
| C/I 6M,1M,BLE         | Adjacent (6 MHz) interference @1Mbps                              |                                                                          | - | -45 | - | dB  |
| C/I CO,2M,BLE         | Co-Channel interference @2Mbps                                    |                                                                          | - | 7   | - | dB  |
| C/I 2M,2M,BLE         | Adjacent (2 MHz) interference @2Mbps                              |                                                                          | - | -3  | - | dB  |
| C/I 4M,2M,BLE         | Adjacent (4 MHz) interference @2Mbps                              |                                                                          | - | -37 | - | dB  |
| C/I 6M,2M,BLE         | Adjacent (6 MHz) interference @2Mbps                              |                                                                          | - | -35 | - | dB  |
| C/I Image,2M,BLE      | Image frequency interference @2Mbps                               |                                                                          | - | -   | - | dB  |
| C/I Image±2M,2M,BLE   | Adjacent (±2 MHz) interference to in-band image frequency         |                                                                          | - | -36 | - | dB  |
| C/I 12M,2M,BLE        | Adjacent (12 MHz) interference @2Mbps                             |                                                                          | - | -35 | - | dB  |
| C/I CO,125K,BLE       | Co-Channel interference @125kbps                                  |                                                                          | - | -6  | - | dB  |
| C/I 1M,125K,BLE       | Adjacent (1 MHz) interference @125kbps                            |                                                                          | - | -13 | - | dB  |
| C/I 2M,125K,BLE       | Adjacent (2 MHz) interference @125kbps                            |                                                                          | - | -37 | - | dB  |
| C/I 3M,125K,BLE       | Adjacent (3MHz) interference @125kbps                             |                                                                          | - | -49 | - | dB  |
| C/I Image,125K,BLE    | Image frequency interference @125kbps                             |                                                                          | - | -31 | - | dB  |
| C/I Image±1M,125K,BLE | Adjacent (±1MHz) interference to in-band image frequency @125kbps |                                                                          | - | -48 | - | dB  |
| C/I CO,500K,BLE       | Co-Channel interference @500kbps                                  |                                                                          | - | 1   | - | dB  |
| C/I 1M,500K,BLE       | Adjacent (1 MHz) interference @500kbps                            |                                                                          | - | -6  | - | dB  |
| C/I 2M,500K,BLE       | Adjacent (2 MHz) interference @500kbps                            |                                                                          | - | -31 | - | dB  |
| C/I 3M,500K,BLE       | Adjacent (3MHz) interference @500kbps                             |                                                                          | - | -43 | - | dB  |
| C/I Image,500K,BLE    | Image frequency interference @500kbps                             |                                                                          | - | -25 | - | dB  |


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|                       |                                                                   |  |   |       |   |    |
|-----------------------|-------------------------------------------------------------------|--|---|-------|---|----|
| C/I Image±1M,500K,BLE | Adjacent (±1MHz) interference to in-band image frequency @500kbps |  | - | -41   | - | dB |
| C/I CO,1M,297         | Co-Channel interference @1Mbps                                    |  | - | TBD - |   | dB |
| C/I 1M,1M,297         | Adjacent (1 MHz) interference @1Mbps                              |  | - | TBD - |   | dB |
| C/I 2M,1M,297         | Adjacent (2 MHz) interference @1Mbps                              |  | - | TBD - |   | dB |
| C/I 3M,1M,297         | Adjacent (3 MHz) interference @1Mbps                              |  | - | TBD - |   | dB |
| C/I Image,1M,297      | Image frequency interference @1Mbps                               |  | - | TBD - |   | dB |
| C/I Image±1M,1M,297   | Adjacent (±1MHz) interference to in-band image frequency @1Mbps   |  | - | TBD - |   | dB |
| C/I 6M,1M,297         | Adjacent (6 MHz) interference @1Mbps                              |  | - | TBD - |   | dB |
| C/I CO,2M,297         | Co-Channel interference @2Mbps                                    |  | - | TBD - |   | dB |
| C/I 2M,2M,297         | Adjacent (2 MHz) interference @2Mbps                              |  | - | TBD - |   | dB |
| C/I 4M,2M,297         | Adjacent (4 MHz) interference @2Mbps                              |  | - | TBD - |   | dB |
| C/I 6M,2M,297         | Adjacent (6 MHz) interference @2Mbps                              |  | - | TBD - |   | dB |
| C/I Image,2M,297      | Image frequency interference @2Mbps                               |  | - | TBD - |   | dB |
| C/I Image±2M,2M,297   | Adjacent (±2MHz) interference to in-band image frequency @2Mbps   |  | - | TBD - |   | dB |
| C/I 12M,2M,297        | Adjacent (12 MHz) interference @2Mbps                             |  | - | TBD - |   | dB |
| C/I CO,250K,297       | Co-Channel interference @250kbps                                  |  | - | TBD - |   | dB |
| C/I 1M,250K,297       | Adjacent (1 MHz) interference @250kbps                            |  | - | TBD - |   | dB |
| C/I 2M,250K,297       | Adjacent (2 MHz) interference @250kbps                            |  | - | TBD - |   | dB |
| C/I 3M,250K,297       | Adjacent (3 MHz) interference @250kbps                            |  | - | TBD - |   | dB |
| C/I Image,250K,297    | Image frequency interference @250kbps                             |  | - | TBD - |   | dB |
| C/I Image±1M,250K,297 | Adjacent (±1MHz) interference to in-band image frequency @250kbps |  | - | TBD - |   | dB |
| C/I CO,1M,N           | Co-Channel interference@1Mbps                                     |  | - | TBD - |   | dB |
| C/I 1M,1M,N           | Adjacent (1 MHz) interference@1Mbps                               |  | - | TBD - |   | dB |
| C/I 2M,1M,N           | Adjacent (2 MHz) interference @1Mbps                              |  | - | TBD - |   | dB |
| C/I 3M,1M,N           | Adjacent (3 MHz) interference @1Mbps                              |  | - | TBD - |   | dB |
| C/I Image,1M,N        | Image frequency interference@1Mbps                                |  | - | TBD - |   | dB |
| C/I Image±1M,1M,N     | Adjacent (±1MHz) interference to in-band image frequency @1Mbps   |  | - | TBD - |   | dB |
| C/I 6M,1M,N           | Adjacent (6 MHz) interference @1Mbps                              |  | - | TBD - |   | dB |
| C/I CO,2M,N           | Co-Channel interference @2Mbps                                    |  | - | TBD - |   | dB |
| C/I 2M,2M,N           | Adjacent (2 MHz) interference @2Mbps                              |  | - | TBD - |   | dB |
| C/I 4M,2M,N           | Adjacent (4 MHz) interference @2Mbps                              |  | - | TBD - |   | dB |




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|                           |                                                                         |  |   |       |  |     |
|---------------------------|-------------------------------------------------------------------------|--|---|-------|--|-----|
| C/I $\gamma$ 6M,2M,N      | Adjacent ( $\gamma$ 6 MHz) interference @2Mbps                          |  | - | TBD - |  | dB  |
| C/I Image,2M,N            | Image frequency interference @2Mbps                                     |  | - | TBD - |  | dB  |
| C/I Image $\pm$ 2M,2M,N   | Adjacent ( $\pm$ 2 MHz) interference to in-band image frequency @2Mbps  |  | - | TBD - |  | dB  |
| C/I $\gamma$ 12M,2M,N     | Adjacent ( $\gamma$ 12 MHz) interference @2Mbps                         |  | - | TBD - |  | dB  |
| C/I CO,250K,N             | Co-Channel interference @250kbps                                        |  | - | TBD - |  | dB  |
| C/I 1M,250K,N             | Adjacent (1 MHz) interference @250kbps                                  |  | - | TBD - |  | dB  |
| C/I 2M,250K,N             | Adjacent (2 MHz) interference @250kbps                                  |  | - | TBD - |  | dB  |
| C/I $\gamma$ 3M,250K,N    | Adjacent ( $\gamma$ 3MHz) interference @250kbps                         |  | - | TBD - |  | dB  |
| C/I Image,250K,N          | Image frequency interference @250kbps                                   |  | - | TBD - |  | dB  |
| C/I Image $\pm$ 1M,250K,N | Adjacent ( $\pm$ 1MHz) interference to in-band image frequency @250kbps |  | - | TBD - |  | dB  |
| C/I CO,250K,B             | Co-Channel interference @250kbps                                        |  | - | TBD - |  | dB  |
| C/I 1M,250K,B             | Adjacent (1MHz) interference @250kbps                                   |  | - | TBD - |  | dB  |
| C/I 2M,250K,B             | Adjacent (2MHz) interference @250kbps                                   |  | - | TBD - |  | dB  |
| C/I $\gamma$ 3M,250K,B    | Adjacent ( $\gamma$ 3MHz) interference @250kbps                         |  | - | TBD - |  | dB  |
| C/I Image,250K,B          | Image frequency interference @250kbps                                   |  | - | TBD - |  | dB  |
| C/I Image $\pm$ 1M,250K,B | Adjacent ( $\pm$ 1MHz) interference to in-band image frequency @250kbps |  | - | TBD - |  | dB  |
| P IMD,5TH,1M              | IMD performance 5 MHz offset @1Mbps                                     |  | - | TBD - |  | dBm |
| P IMD,5TH,2M              | IMD performance 10 MHz offset @2Mbps                                    |  | - | TBD - |  | dBm |

Table 5-11 RSSI characteristics

| Symbol  | Description           | Conditions | Parameter |         |     | Unit |
|---------|-----------------------|------------|-----------|---------|-----|------|
|         |                       |            | Min       | Type    | Max |      |
| RSSIRFC | RSSI indication range |            | -90       | -       | -15 | dBm  |
| RSSIAuu | RSSI accuracy         |            | -         | $\pm$ 2 | -   | dB   |
| RSSIRes | RSSI resolution       |            | -         | 0.25    | -   | dB   |
| RSSIPer | RSSI Sample period    |            | -         | 0.25    | -   | us   |

Table 5-12 RF Timing Characteristics

| Symbol  | Description                                                                              | Conditions | Parameter |     |       | Unit |
|---------|------------------------------------------------------------------------------------------|------------|-----------|-----|-------|------|
|         |                                                                                          |            | Min       | Typ | Max   |      |
| TOSC,EN | Crystal oscillator settling time                                                         |            | -         |     | TBD - | us   |
| TTX,EN  | Time between TXEN task and READY event after channel FREQUENCY configured                |            | -         |     | TBD - | us   |
| TRX,EN  | Time between the RXEN task and READY event after channel FREQUENCY configured in default |            | -         |     | TBD - | us   |


**PAN3740 Series Sub-1G & 2.4G Multi-mode Wireless SOC Chip**

|             |                                                                       |  |   |       |   |    |
|-------------|-----------------------------------------------------------------------|--|---|-------|---|----|
|             | mode                                                                  |  |   |       |   |    |
| TTX,DISABLE | Time between DISABLE task and DISABLED event when the radio was in TX |  | - | TBD - |   | us |
| TRX,DISABLE | Time between DISABLE task and DISABLED event when the radio was in RX |  | - | TBD - |   | us |
| TRX-TX      | The time taken to switch from RX to TX or TX to RX                    |  | - | 150   | - | us |

Table 5-13 RF power consumption characteristics

| Symbol           | Description                       | Conditions | Parameter |       |     | Unit |
|------------------|-----------------------------------|------------|-----------|-------|-----|------|
|                  |                                   |            | Min       | Type  | Max |      |
| ITX,P9dBm,DCDC   | TX only run current 9dBm @DC-DC   |            | -         | 13.97 | -   | mA   |
| ITX,P6dBm,DCDC   | TX only run current 6dBm @DC-DC   |            | -         | 9.59  | -   | mA   |
| ITX,P4dBm,DCDC   | TX only run current 4dBm @DC-DC   |            | -         | 6.67  | -   | mA   |
| ITX,P0dBm,DCDC   | TX only run current 0dBm @DC-DC   |            | -         | 5.06  | -   | mA   |
| ITX,P-4dBm,DCDC  | TX only run current -4dBm @DC-DC  |            | -         | 2.56  | -   | mA   |
| ITX,P-8dBm,DCDC  | TX only run current -8dBm @DC-DC  |            | -         | 2.15  | -   | mA   |
| ITX,P-12dBm,DCDC | TX only run current -12dBm @DC-DC |            | -         | 1.87  | -   | mA   |
| ITX,P-16dBm,DCDC | TX only run current -16dBm @DC-DC |            | -         | 1.64  | -   | mA   |
| ITX,P-20dBm,DCDC | TX only run current -20dBm @DC-DC |            | -         | 1.53  | -   | mA   |
| ITX,P-40dBm,DCDC | TX only run current -40dBm @DC-DC |            | -         | 1.12  | -   | mA   |
| ITX,P9dBm,LDO    | TX only run current 9dBm @LDO     |            | -         | 35.55 | -   | mA   |
| ITX,P6dBm,LDO    | TX only run current 6dBm @LDO     |            | -         | 16.28 | -   | mA   |
| ITX,P4dBm,LDO    | TX only run current 4dBm @LDO     |            | -         | 12.1  | -   | mA   |
| ITX,P0dBm,LDO    | TX only run current 0dBm @LDO     |            | -         | 9.43  | -   | mA   |
| ITX,P-4dBm,LDO   | TX only run current -4dBm @LDO    |            | -         | 5.74  | -   | mA   |
| ITX,P-8dBm,LDO   | TX only run current -8dBm @LDO    |            | -         | 5.08  | -   | mA   |
| ITX,P-12dBm,LDO  | TX only run current -12dBm @LDO   |            | -         | 4.62  | -   | mA   |
| ITX,P-16dBm,LDO  | TX only run current -16dBm @LDO   |            | -         | 4.04  | -   | mA   |
| ITX,P-20dBm,LDO  | TX only run current -20dBm @LDO   |            | -         | 3.92  | -   | mA   |
| ITX,P-40dBm,LDO  | TX only run current -40dBm @LDO   |            | -         | 3.00  | -   | mA   |
| IRX,1M,DCDC      | RX 1Mbps current @DC-DC           |            | -         | 2.50  | -   | mA   |
| IRX,2M,DCDC      | RX 2Mbps current @DC-DC           |            | -         | 2.71  | -   | mA   |
| IRX,1M,LDO       | RX 1Mbps current @LDO             |            | -         | 4.82  | -   | mA   |
| IRX,2M,LDO       | RX 2Mbps current @LDO             |            | -         | 5.67  | -   | mA   |



## Test conditions and methods.

1. Transceiver power consumption tested in 1M mode using BLE ADV broadcast mode.
2. 2M mode is used as the power consumption when BLE connection.
3. The power consumption tested is the RF peak power.
4. The test method uses the total power consumption minus the power consumption of the MCU when the RF is not operating to calculate the final power consumption.
5. The sample software tested is based on peripheral\_hr.
6. 3.3V Power Supply

## 5.2.4 GPIO Features

Table 5-14 Single IO combination test

| Symbol    | Description                                                   | Conditions                      | Parameter |      |         | Unit |
|-----------|---------------------------------------------------------------|---------------------------------|-----------|------|---------|------|
|           |                                                               |                                 | Min       | Type | Max     |      |
| VIH       | Input high voltage                                            | TA=25°C                         | 0.7*VDD - |      | VDD     | V    |
| VIL       | Input low voltage                                             | Load capacity =20pF,<br>TA=25°C | VSS       | -    | 0.2*VDD | V    |
| VHYS      | Input hysteresis voltage,<br>Vhys=VIH-VIL                     | TA=25°C                         | -         | -    | 0.3*VDD | V    |
| Clana     | Analog input capacitors                                       | TA=25°                          | -         | 300  | -       | f    |
| ILkg      | Leakage current, open-drain mode<br>or input mode             | VDD,VIN=3.6V -                  |           | -    | TBD     | uA   |
| RPU       | Pull-up resistor                                              | Vin = VSS,<br>VDD =3.3V         | -         | 50   | -       | kΩ   |
| RPD       | Pull-down resistor                                            | Vin = VSS,<br>VDD =3.3V         | -         | 100  | -       | kΩ   |
| VI        | Input voltage                                                 | TA=25°C                         | VSS       | -    | VDD     | V    |
| VO        | Output voltage                                                | TA=25°C                         | VSS       | -    | VDD     | V    |
| Isource   | Source current<br>(Push-pull output)<br>(Except for P03, P15) | Vin = VDD-0.5V                  | 8.5       | 9.5  | 10.5    | mA   |
|           | Source current<br>(Push-pull output)<br>(P03, P15)            |                                 |           |      |         | mA   |
| ISink     | Sink current(Push-pull output)                                | Vin = VSS + 0.5V,<br>TA = 25°C  | 19.5      | 20.5 | 21.5    | mA   |
| fPort_CLK | IO output frequency                                           | Load capacitance =<br>20pF      | -         | -    | 48      | MHz  |

Table 5-15 Combination test

| Description                     | Conditions        | Status                                                                                                 | Remark |
|---------------------------------|-------------------|--------------------------------------------------------------------------------------------------------|--------|
| IO default state after power on | VDD=3.3V ,TA=25°C | P00, P01: Pull-up input state<br>Others: High resistance state                                         |        |
| IO status in deepsleep mode     | VDD=3.3V ,TA=25°C | Deepsleep: All gpio retention<br>Standby_m1: All gpio retention<br>Standby_m0: P00, P01, P02 retention |        |



|                    |                   |                                                                |  |
|--------------------|-------------------|----------------------------------------------------------------|--|
| IO status at reset | VDD=3.3V ,TA=25°C | P00, P01: Pull-up input state<br>Others: High resistance state |  |
|--------------------|-------------------|----------------------------------------------------------------|--|

Table 5-16 nRESET input characteristics

| Symbol     | Description                          | Conditions                      | Parameter |      |            | Unit       |
|------------|--------------------------------------|---------------------------------|-----------|------|------------|------------|
|            |                                      |                                 | Min       | Type | Max        |            |
| VILR       | Negative threshold voltage, nRESET   | VDD=1.8V-3.3V ,TA=25°C          | -         | -    | 0.22*VDD V |            |
| VIHR       | Positive threshold voltage, nRESET   | VDD=1.8V-3.3V ,TA=25°C 0.48*VDD |           |      | -          | V          |
| Vhys_rst   | Schmitt Trigger Voltage Hysteresis   | VDD=1.8V-3.3V ,TA=25°C          | -         | -    | 0.26*VDD V |            |
| RRST       | nRESET pin internal pull-up resistor | VDD=3.3V ,TA=25°C               | -         | 7.5  | -          | k $\Omega$ |
| tFR, 0.3pF | nRESET pin input filter pulse time   | VDD=3.3V ,TA=25°C               | -         | TBD  | -          | ns         |

### 5.2.5 Reset Characteristics

Table 5-17 Reset characteristics

| Symbol   | Description                           | Conditions                                             | Parameter |      |        | Unit       |
|----------|---------------------------------------|--------------------------------------------------------|-----------|------|--------|------------|
|          |                                       |                                                        | Min       | Typ  | Max    |            |
| VBOD     | Brown-out detection voltage threshold | BODSEL<2:0> = 000(falling edge)<br>dVDD/dt $\geq$ 3V/s | -         | 1.84 | -      | V          |
|          |                                       | BODSEL<2:0> = 001(falling edge)<br>dVDD/dt $\geq$ 3V/s | -         | 1.96 | -      |            |
|          |                                       | BODSEL<2:0> = 010(falling edge)<br>dVDD/dt $\geq$ 3V/s | -         | 2.05 | -      |            |
|          |                                       | BODSEL<2:0> = 011(falling edge)<br>dVDD/dt $\geq$ 3V/s | -         | 2.16 | -      |            |
|          |                                       | BODSEL<2:0> = 100(falling edge)<br>dVDD/dt $\geq$ 3V/s | -         | 2.25 | -      |            |
|          |                                       | BODSEL<2:0> = 101(falling edge)<br>dVDD/dt $\geq$ 3V/s | -         | 2.35 | -      |            |
|          |                                       | BODSEL<2:0> = 110(falling edge)<br>dVDD/dt $\geq$ 3V/s | -         | 2.47 | -      |            |
| VBODhys  | BOD hysteresis voltage                | dVDD/dt $\geq$ 3V/s                                    | 100       | -    | 160 mV |            |
| TBOD_RE1 | BOD response Time(Normal mode)        | dVDD/dt $\geq$ 3V/s                                    | 1         | 32   | 32     | Slow clock |
| IBOD     | BOD operating current                 | dVDD/dt $\geq$ 3V/s                                    | -         | 0.5  | -      | $\mu$ A    |
| VPOR     | Power on reset voltage threshold      | Rising edge, dvdd/dt $\geq$ 3V/s                       | -         | 1.71 | -      | V          |
|          |                                       | Falling edge, dvdd/dt $\geq$ 3V/s                      | -         | 1.69 | -      | V          |
| TPOR     | POR settling time                     | VBAT =3.3V                                             | -         | 1.12 | -      | ms         |





|          |                                 |                            |   |      |    |            |
|----------|---------------------------------|----------------------------|---|------|----|------------|
| VLVR     | LVR detection voltage threshold | Falling edge, dvdd/dt 3V/s | - | 1.85 | -  | V          |
| TLVR_REF | LVR response time               | TA=25°C, dVDD/dt 3V/s      | 1 | 32   | 32 | Slow clock |
| ILVR     | LVR operating current           | TA=25°C, dVDD/dt 3V/s      | - | 0.1  | -  | uA         |

## 5.2.6 Clock Characteristics

Table 5-18 HXTAL characteristics

| Symbol    | Description                                     | Conditions                            | Parameter |     |     | Unit |
|-----------|-------------------------------------------------|---------------------------------------|-----------|-----|-----|------|
|           |                                                 |                                       | Min       | Typ | Max |      |
| HkDJ      | High speed crystal oscillator (HXTAL) frequency | VDD=3.3V, TA=25°C                     | -         | 32  | -   | MHz  |
| CLoadHXTL | Crystal load capacitance                        | VDD=3.3V, TA=25°C                     | 7         | 9   | 12  | pF   |
| IDX       | HXTAL oscillator operating current              | VDD=3.3V, TA=25°C                     | -         | 250 | -   | μA   |
| tX        | HXTAL oscillator startup time                   | VDD=3.3V, TA=25°C, ESR=40Ω, CHXTL=9pF | -         | 465 | -   | μs   |
| tX Quick  | HXTAL oscillator Quick startup time             | VDD=3.3V, TA=25°C, ESR=40Ω, CHXTL=9pF | -         | 155 | -   | μs   |
| ESR       |                                                 |                                       | -         | 40  | -   | Ω    |
| FTOLHXTL  | Frequency tolerance for the crystal             | VDD=3.3V, TA=25°C                     | -20       | -   | 20  | ppm  |
| PDHX      | Drive level                                     | VDD=3.3V, TA=25°C                     | -         | -   | 100 | μW   |

Table 5-19 LXTAL characteristics

| Symbol        | Description                                      | Conditions            | Parameter |        |         | Unit |
|---------------|--------------------------------------------------|-----------------------|-----------|--------|---------|------|
|               |                                                  |                       | Min       | Type   | Max     |      |
| FWf           | Low speed crystal oscillator (LXTAL) frequency   | VDD=3.3V, TA=25°C     |           | 32.768 | -       | kHz  |
| IDDLXTL       | LXTAL oscillator operating current               | VDD=3.3V, TA=25°C 0.3 |           | 0.45   | 0.67    | μA   |
| tXTL          | LXTAL oscillator Normal startup time             | VDD=3.3V, TA=25°C 330 |           | 550    | 1500 ms |      |
| tSULXTL Quick | LXTAL oscillator Quick startup time              | VDD=3.3V, TA=25°C     |           | 125    | -       | ms   |
| ESRLXTL       | Equivalent series resistance<br>6 pF < CL < 9 pF | VDD=3.3V, TA=25°C     |           | 70     | -       | kΩ   |
| CLoadLXTL     | Crystal load capacitance                         | VDD=3.3V, TA=25°C     |           | 7      | -       | pF   |
| PDLXTL        | Drive level                                      | VDD=3.3V, TA=25°C     |           | -      | 1       | μW   |


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Table 5-20 32MHz RCH characteristics

| Symbol    | Description                                        | Conditions                       | Parameter |      |      | Unit |
|-----------|----------------------------------------------------|----------------------------------|-----------|------|------|------|
|           |                                                    |                                  | Min       | Type | Max  |      |
| fIRC32M   | Crystal frequency                                  | VDD=3.3V, TA=25°C                | -         | 32   | -    | MHz  |
| ACCIRC32M | Frequency accuracy                                 | VDD=3.3V, TA=-40°C ~ +105°C -    | -         | -    | -    | %    |
|           |                                                    | VDD=3.3V, TA=-20°C ~ +85°C -     | -         | -    | -    | %    |
|           |                                                    | VDD=3.3V, TA=25°C                | -         | ±1   | -    | %    |
| DIRC32M   | IRC32M oscillator duty cycle                       | VDD=3.3V, fIRC32M=32MHz, TA=25°C | 48        | 50   | 52 % |      |
| IDDIRC32M | Operating current                                  | VDD=3.3V, fIRC8M=32MHz, TA=25°C  | -         | 82   | -    | μA   |
| tSUIRC32M | Startup time                                       | VDD=3.3V, fIRC32M=32MHz, TA=25°C | -         | 5    | -    | μs   |
| dfIRC32M  | 25°C, the frequency drifts with the supply voltage | VDD=1.8V~3.6V, TA=25°C           | -         | 0.5  | -    | %/V  |

Table 5-21 32kHz RCL characteristics

| Symbol    | Description                                        | Conditions                            | Parameter |      |     | Unit |
|-----------|----------------------------------------------------|---------------------------------------|-----------|------|-----|------|
|           |                                                    |                                       | Min       | Type | Max |      |
| fIRC32k   | Crystal frequency                                  | VDD=3.3V, TA=25°C                     | -         | 32   | -   | kHz  |
| ACCIRC32K | Frequency accuracy                                 | VDD=3.3V, TA=25°C (After calibration) | -         | ±500 | -   | ppm  |
| DIRC32K   | IRC32K oscillator duty cycle                       | VDD=3.3V, fIRC32K=32kHz, TA=25°C      | 48        | 50   | 52  | %    |
| IDDIRC32K | Operating current                                  | VDD=3.3V, fIRC8K=32kHz, TA=25°C       | -         | 310  | -   | nA   |
| tSUIRC32K | Startup time                                       | VDD=3.3V, fIRC32K=32kHz, TA=25°C      | -         | 480  | -   | μs   |
| dfIRC32K  | 25°C, the frequency drifts with the supply voltage | VDD=1.8V~3.6V, TA=25°C                | -         | 1    | -   | %/V  |

Table 5-22 DPLL characteristics

| Symbol | Description               | Conditions                          | Parameter |      |     | Unit |
|--------|---------------------------|-------------------------------------|-----------|------|-----|------|
|        |                           |                                     | Min       | Type | Max |      |
| f      | PLL input clock frequency | VDD=3.3V, TA=25°C PLL output        | -         | 32   | -   | MHz  |
| fPLL   | clock frequency           | VDD=3.3V, TA=25°C Operating current | 48        | 48   | 48  | MHz  |
| IPLL   |                           | VDD=3.3V, TA=25°C                   | 96        | 105  | 145 | μA   |



## 5.2.7 ADC Characteristics

Table 5-23 Power supply and input range conditions

| Symbol                                | Description                                                               | Conditions                                   | Parameter |      |            | Unit |
|---------------------------------------|---------------------------------------------------------------------------|----------------------------------------------|-----------|------|------------|------|
|                                       |                                                                           |                                              | Min       | Type | Max        |      |
| V <sub>Ax</sub> (V <sub>BG</sub> adc) | Analog input voltage range, VBG (1.2V)                                    | VDD=3.3V, TA=25°C                            | 0         | -    | 1.2        | V    |
| V <sub>Ax</sub> (VDD)                 | Analog input voltage range, VDD                                           | VDD=3.3V, TA=25°C                            | 0         | -    | VDD V      |      |
| I <sub>ADC</sub>                      | ADC supply current                                                        | VDD=3.3V, TA=25°C<br>F <sub>adc</sub> =16MHz | -         | 0.55 | -          | mA   |
| C <sub>sample</sub>                   | Internal sample and hold capacitors (PAD and PCB capacitors not included) |                                              | -         | 10   | -          | pF   |
| R <sub>ADC</sub>                      | Sampling switch resistance 0V ~ V <sub>Ax</sub> ~ VDD                     |                                              | -         | 300  | -          | Ω    |
| R <sub>In</sub>                       | External input impedance, continuous sampling                             | 0V ~ V <sub>Ax</sub> ~ VDD                   | 0.86      | -    | 4734.70 kΩ |      |

Table 5-24 ADC built-in voltage reference

| Symbol              | Description                     | Conditions                      | Parameter |      |     | Unit   |
|---------------------|---------------------------------|---------------------------------|-----------|------|-----|--------|
|                     |                                 |                                 | Min       | Type | Max |        |
| V <sub>BG</sub> ADC | Internal 1.2V Reference Voltage | VDD=3.3V, TA=25°C TA=-40~-105°C | 1.1       | 1.2  | 1.3 | V      |
| T <sub>Coef</sub>   | Temperature factor              | VDD=1.8V~3.6V                   | -         | 30   | -   | ppm/°C |

Table 5-25 Time parameters

| Symbol            | Description         | Conditions        | Parameter |      |      | Unit               |
|-------------------|---------------------|-------------------|-----------|------|------|--------------------|
|                   |                     |                   | Min       | Type | Max  |                    |
| F <sub>ADC</sub>  | ADC clock frequency | VDD=3.3V, TA=25°C | 4         | 16   |      | MHz                |
| T <sub>S</sub>    | Sample time         | VDD=3.3V, TA=25°C | 4         | 1539 | 8192 | 1/F <sub>adc</sub> |
| T <sub>CONV</sub> | Conversion time     | VDD=3.3V, TA=25°C | 32        | 1580 | 8298 | 1/F <sub>adc</sub> |

Table 5-26 Linearity parameters

| Symbol | Description                  | Conditions        | Parameter |      |     | Unit |
|--------|------------------------------|-------------------|-----------|------|-----|------|
|        |                              |                   | Min       | Type | Max |      |
| INL    | Integral linearity error     | VDD=3.3V, TA=25°C | -         | -    | ±3  | LSB  |
| DNL    | Differential linearity error | VDD=3.3V, TA=25°C | -         | -    | ±2  | LSB  |


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|      |                                    |                                                         |   |       |   |     |
|------|------------------------------------|---------------------------------------------------------|---|-------|---|-----|
| SNR  | Signal to Noise Ratio              | Fadc = 16MHz<br>Input Clock 250kHz<br>VDD=3.3V, TA=25°C | - | 64.3  | - | dB  |
| THD  | Total harmonic distortion          |                                                         | - | 75    | - | dB  |
| SFDR | Spurious-free signal dynamic range |                                                         | - | 77.29 | - | dB  |
| ENOB | Effective number of bits           |                                                         | - | 10.33 | - | Bit |

Table 5-27 RIN

| ADC significant bit FADC(MHz) |    | Ts(cycles) | Ts(us) | Rinmax(k $\Omega$ ) |
|-------------------------------|----|------------|--------|---------------------|
| 12                            | 32 | 4          | 0.125  | 0.86                |
| 12                            | 32 | 8          | 0.25   | 2.01                |
| 12                            | 32 | 32         | 1      | 8.95                |
| 12                            | 32 | 64         | 2      | 18.20               |
| 12                            | 32 | 128        | 4      | 36.69               |
| 12                            | 32 | 8192       | 256    | 2367.20             |
| 12                            | 16 | 4          | 0.25   | 2.01                |
| 12                            | 16 | 8          | 0.5    | 4.32                |
| 12                            | 16 | 32         | 2      | 18.20               |
| 12                            | 16 | 64         | 4      | 36.69               |
| 12                            | 16 | 128        | 8      | 73.68               |
| 12                            | 16 | 8192       | 512    | 4734.70             |

Note: The sampling condition is continuous sampling.

### 5.2.8 PMU Characteristics

Table 5-28 PMU characteristics

| Symbol  | Description                                           | Conditions        | Parameter |     |       | Unit |
|---------|-------------------------------------------------------|-------------------|-----------|-----|-------|------|
|         |                                                       |                   | Min       | Typ | Max   |      |
| VDJ     | HLDO output voltage range, external capacitor         | VDD=3.3V, TA=25°C | 1.1       | 1.2 | 1.4 V |      |
| VDDPSRR | Power supply rejection ratio of VDD VDD=3.3V, TA=25°C |                   | -15       | -   | -     | dB   |

### 5.2.9 DCDC Characteristics

Table 5-29 DCDC characteristics

| Symbol    | Description          | Conditions                                          | Parameter |     |       | Unit |
|-----------|----------------------|-----------------------------------------------------|-----------|-----|-------|------|
|           |                      |                                                     | Min       | Typ | Max   |      |
| VIN_DCDC  | Input voltage range  | VDD=3.3V, TA=25°C                                   | 2         | -   | 3.6 V |      |
| VOUT_DCDC | Output voltage range | VDD=3.3V, TA=25°C                                   | 1.2       | 1.5 | 2     | V    |
| TEN_DCDC  | Standup time         | VDD=3.3V, ILOAD=10mA,<br>TA=25°C, LDCR=80m $\Omega$ | -         | 200 | -     | us   |





|           |                         |                                                     |   |     |   |          |
|-----------|-------------------------|-----------------------------------------------------|---|-----|---|----------|
| $\eta$    | Efficiency              | VDD=3.3V, ILOAD=10mA,<br>TA=25°C, LDCR=80m $\Omega$ | - | 88  | - | %        |
| VRPLDCDC  | Ripple                  | VDD=3.3V, TA=25°C                                   | - | 43  | - | mv       |
| IOUT      | Drive peak current      | VDD=3.3V, TA=25°C                                   | - | 100 | - | mA       |
| IAVG      | Drive average current   | VDD=3.3V, TA=25°C                                   | - | 30  | - | mA       |
| LDCDC     | Effective inductance    | VDD=3.3V, TA=25°C                                   | - | 2.2 | - | $\mu$ H  |
| COUT DCDC | Effective load capacity | VDD=3.3V, TA=25°C VDD=3.3V,                         | 1 | 4.7 | - | $\mu$ F  |
| Fosc DCDC | Oscillation frequency   | TA=25°C                                             | - | -   | - | 1000 kHz |

## 5.2.10 MCU Current Characteristics

| Symbol   | Parameter                                           | Conditions                                                           |     | DCDC OFF DCDC ON |         |
|----------|-----------------------------------------------------|----------------------------------------------------------------------|-----|------------------|---------|
|          |                                                     |                                                                      |     | Typ(mA)          | Typ(mA) |
| Run mode | All peripherals clock on,<br>run while(1) in flash  | System clock source:<br>RCH<br>(.cal 32M)                            | 4M  | 0.91             | 0.686   |
|          |                                                     |                                                                      | 8M  | 1.3              | 0.916   |
|          |                                                     |                                                                      | 16M | 2.09             | 1.36    |
|          |                                                     |                                                                      | 32M | 3.77             | 2.28    |
|          |                                                     | System clock source:<br>XTH<br>(.off rch)                            | 4M  | 1.42             | 0.985   |
|          |                                                     |                                                                      | 8M  | 1.81             | 1.2     |
|          |                                                     |                                                                      | 16M | 2.61             | 1.64    |
|          |                                                     |                                                                      | 32M | 4.29             | 2.54    |
|          |                                                     | System clock source:<br>DPLL<br>(.ref rch(.cal 32M))<br>(.base 48M)  | 6M  | 1.56             | 1.05    |
|          |                                                     |                                                                      | 12M | 2.15             | 1.38    |
|          |                                                     |                                                                      | 24M | 3.4              | 2.05    |
|          |                                                     |                                                                      | 48M | 5.71             | 3.44    |
|          |                                                     | System clock source:<br>DPLL<br>(.ref xth)<br>(.base 48M) (.off rch) | 6M  | 2.08             | 1.34    |
|          |                                                     |                                                                      | 12M | 2.69             | 1.67    |
|          |                                                     |                                                                      | 24M | 3.95             | 2.34    |
|          |                                                     |                                                                      | 48M | 6.18             | 3.57    |
|          |                                                     | System clock source:<br>DPLL<br>(.ref rch(.cal 32M))<br>(.base 64M)  | 4M  | 1.39             | 0.953   |
|          |                                                     |                                                                      | 8M  | 1.78             | 1.17    |
|          |                                                     |                                                                      | 16M | 2.58             | 1.61    |
|          |                                                     |                                                                      | 32M | 4.27             | 2.51    |
|          |                                                     | System clock source:<br>DPLL<br>(.ref xth)<br>(.base 64M) (.off rch) | 4M  | 1.9              | 1.25    |
|          |                                                     |                                                                      | 8M  | 2.3              | 1.46    |
|          |                                                     |                                                                      | 16M | 3.13             | 1.91    |
|          |                                                     |                                                                      | 32M | 4.85             | 2.82    |
|          | All peripherals clock off,<br>run while(1) in flash | System clock source:<br>RCH<br>(.cal 32M)                            | 4M  | 0.736            | 0.523   |
|          |                                                     |                                                                      | 8M  | 0.943            | 0.719   |
|          |                                                     |                                                                      | 16M | 1.37             | 0.963   |
|          |                                                     |                                                                      | 32M | 2.23             | 1.43    |
|          |                                                     | System clock source:<br>XTH<br>(.off rch)                            | 4M  | 0.945            | 0.723   |
|          |                                                     |                                                                      | 8M  | 1.15             | 0.84    |
|          |                                                     |                                                                      | 16M | 1.57             | 1.07    |
|          |                                                     |                                                                      |     |                  |         |


**PAN3740 Series Sub-1G & 2.4G Multi-mode Wireless SOC Chip**

|  |  |                                                                      |     |       |       |
|--|--|----------------------------------------------------------------------|-----|-------|-------|
|  |  | System clock source:<br>DPLL<br>(.ref rch(.cal 32M))<br>(.base 48M)  | 32M | 2.42  | 1.54  |
|  |  |                                                                      | 6M  | 1.01  | 0.747 |
|  |  |                                                                      | 12M | 1.32  | 0.921 |
|  |  |                                                                      | 24M | 1.96  | 1.25  |
|  |  | System clock source:<br>DPLL<br>(.ref xth)<br>(.base 48M) (.off rch) | 48M | 3.29  | 2.00  |
|  |  |                                                                      | 6M  | 1.22  | 0.903 |
|  |  |                                                                      | 12M | 1.54  | 1.05  |
|  |  |                                                                      | 24M | 2.19  | 1.41  |
|  |  |                                                                      | 48M | 3.54  | 2.12  |
|  |  | System clock source:<br>DPLL<br>(.ref rch(.cal 32M))<br>(.base 64M)  | 4M  | 0.925 | 0.693 |
|  |  |                                                                      | 8M  | 1.14  | 0.818 |
|  |  |                                                                      | 16M | 1.56  | 1.05  |
|  |  |                                                                      | 32M | 2.43  | 1.53  |
|  |  | System clock source:<br>DPLL<br>(.ref xth)<br>(.base 64M) (.off rch) | 4M  | 1.14  | 0.831 |
|  |  |                                                                      | 8M  | 1.35  | 0.947 |
|  |  |                                                                      | 16M | 1.78  | 1.18  |
|  |  |                                                                      | 32M | 2.66  | 1.66  |

Notes: Test conditions - DVDD=1.2V, VDD=3.3V, VBG=1.2V, TA=25°C

| Symbol     | Parameter                | Conditions       |            | Typ(µA)       |
|------------|--------------------------|------------------|------------|---------------|
|            | standby m0 wake by gpio  |                  |            | 0.28          |
| Symbol     | Parameter                | Conditions       | Clk source | Typ(µA)       |
| standby m1 | wake by 32k timer        | sw=0x0d (Note 1) | XTL 1.88   | (Note 3)      |
|            |                          | sw=0x1f (Note 1) | XTL 2.18   | (Note 3)      |
|            | wake by gpio             | sw=0x0d (Note 1) |            | 1.32(Note 3)  |
|            |                          | sw=0x1f (Note 1) |            | 1.63(Note 3)  |
|            | wake by wdt              | sw=0x0d (Note 1) | XTL 1.85   | (Note 3)      |
|            |                          | sw=0x1f (Note 1) | XTL 2.18   | (Note 3)      |
|            | wake by lvr              | sw=0x0d (Note 1) |            | 2.05(Note 3)  |
|            |                          | sw=0x1f (Note 1) |            | 2.42(Note 3)  |
| Deepsleep  | wake by 32k timer        | sw=0x0d (Note 1) | XTL 2.96   | (Note 3)      |
|            |                          | sw=0x1f (Note 1) | XTL 3.37   | (Note 3)      |
|            | wake by gpio             | sw=0x0d (Note 1) |            | 2.46 (Note 3) |
|            |                          | sw=0x1f (Note 1) |            | 2.86(Note 3)  |
|            | wake by wdt              | sw=0x0d (Note 1) | XTL 2.99   | (Note 3)      |
|            |                          | sw=0x1f (Note 1) | XTL 3.37   | (Note 3)      |
|            | wake by peripheral timer | sw=0x0d (Note 1) | XTL 4.03   | (Note 3&4)    |
|            |                          | sw=0x1f (Note 1) | XTL 4.73   | (Note 3)      |
|            | wake by 32k timer        | sw=0x0d (Note 1) | XTL* 4.06  | (Note 3&4)    |
|            |                          | sw=0x1f (Note 1) | XTL* 4.69  | (Note 3)      |



|       |              |                                          |      |
|-------|--------------|------------------------------------------|------|
| Sleep | wake by gpio | LDO all peripheral clocks on             | 4620 |
|       |              | LDO non-essential peripheral clocks off  | 1820 |
|       |              | DCDC all peripheral clocks on            | 2700 |
|       |              | DCDC non-essential peripheral clocks off | 1210 |

Note: Test conditions - DVDD=1.2V, VDD=3.3V, VBG=1.2V, TA=25°C, DCDC ON

Note\*: PWM output enable.

Notes:

1. The **sw** indicates the module control switch for PAN107x power retention in low-power mode, bit 1 for power retention, bit 0 for power down, 0d = sram32k+phy\_sram+cpu\_retention, 1f = all\_sram\_retention
2. In LPLDOH Enhance mode, the *lpldoh* voltage is affected by *lpldoh* trim and *vref* trim, where *vref* trim compensates *lpldoh* when *lpldoh* jitter, so that it does not fall to the voltage of *vref* trim as much as possible.
3. In LPLDOH mode2, the *lpldoh* trim voltage does not take effect, and the effective voltage is the *vref\_trim* voltage. The *lpldoh* undershoot does not occur in this mode, and the bottom current is reduced.
4. PWM can output waveforms normally in *Deepsleep* mode, but only in *Deepsleep* mode2, and see Note 5 for instructions on *Deepsleep* mode2.
5. When the register *LP\_FL\_CTRL* [31] is 1, the switch between LPLDOL and LPLDOH is connected, and LPLDOL enable is off. LPLDOH supplies power to LPLDOL. This is the mode 2. Mode 2 power consumption increases.

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■ ■ ■

Figure 6-1 Reference schematic diagram



7 Packaging Information

7.1 QFN48 Package

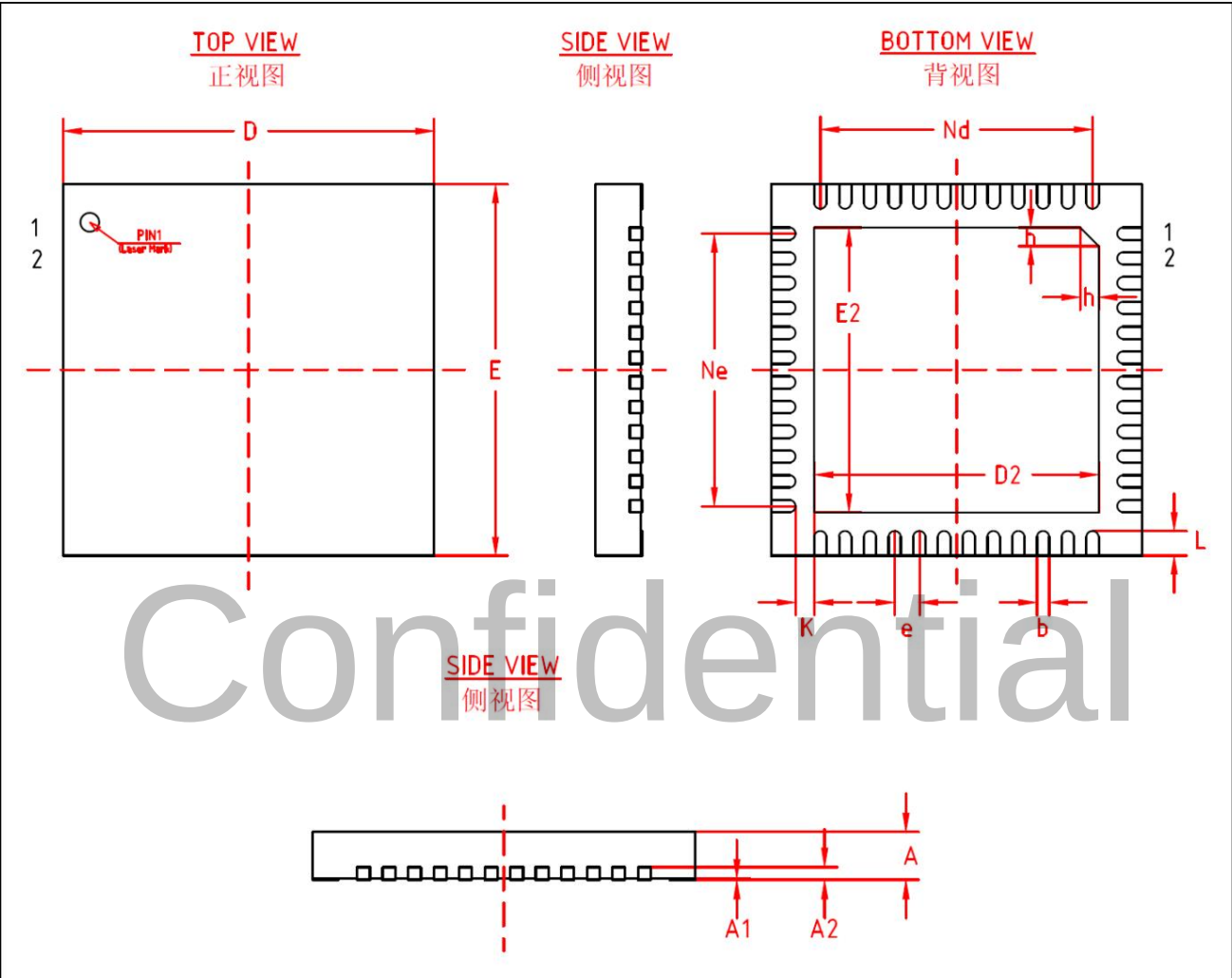


Figure 7-1 Packaging diagram

Table 7-1 Package dimensions

| SYMBOL | MIN (mm)  | NOM (mm) | MAX (mm) |
|--------|-----------|----------|----------|
| A      | 0.70      | 0.75     | 0.80     |
| A1     | -         | 0.02     | 0.05     |
| A2     | 0.203 REF |          |          |
| b      | 0.15      | 0.20     | 0.25     |
| D      | 5.90      | 6.00     | 6.10     |
| D2     | 4.55      | 4.60     | 4.65     |
| E      | 5.90      | 6.00     | 6.10     |
| E2     | 4.55      | 4.60     | 4.65     |
| e      | 0.40 BSC  |          |          |
| K      | 0.25      | 0.30     | 0.35     |



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|    |          |      |      |
|----|----------|------|------|
| L  | 0.35     | 0.40 | 0.45 |
| h  | 0.25     | 0.30 | 0.35 |
| Ne | 4.40 BSC |      |      |
| Nd | 4.40 BSC |      |      |

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## Abbreviations

|                                                      |                                                  |
|------------------------------------------------------|--------------------------------------------------|
| ACK Acknowledgement signal                           | MCU Microcontroller Unit                         |
| ADC Analog-to-Digital Converter                      | MISO Master In Slave Out                         |
| bandgap bandgap reference                            | MM Discharge mode for powered machines           |
| BLE Bluetooth Low Energy                             | MOSI Master Out Slave In                         |
| BOD brownout detection                               | PLL Phase Locked Loop                            |
| BQB Bluetooth Certification                          | POR Power-On Reset                               |
| Cache                                                | PWM Pulse Width Modulation                       |
| CAD Channel Activity Detection                       | QDEC Quadrature Decoder                          |
| CDM Charged Device Model                             | RAM Random Access Memory                         |
| Chirp Linear Frequency Modulation                    | trg trigger                                      |
| CS Chip Select                                       | OSC Oscillator                                   |
| CTS Receive Request                                  | PA Power Amplifier                               |
| DAC Digital to Analog Converter                      | RC Resistor Capacitor Crystal Oscillator         |
| DCDC Step-Down DC Converter                          | RF                                               |
| DMA Direct Memory Access                             | PLL Phase Locked Loop                            |
| DPLL Digital Phase-Locked Loop                       | RSSI Received Signal Strength                    |
| eFuse One-Time Programmable Memory                   | RX                                               |
| ECC elliptic cryptography                            | SDA data signal line                             |
| ESD Electrostatic Discharge                          | SF Spreading Factor                              |
| ext External IO port                                 | SPI Serial Peripheral Interface                  |
| ETSI European Telecommunications Standards Institute | SRAM Static Random Access Memory                 |
| FCC Federal Communications Commission                | SWD Serial Wire Debug                            |
| FMC Flash controller                                 | STB standby mode                                 |
| FIFO First In First Out                              | TEMP Temperature Sensor                          |
| GPIO General Purpose I/O.                            | TX                                               |
| HBM discharge pattern of a charged human body        | RTS Send Request                                 |
| HLDO Strong Drive Low Dropout Linear Regulator       | UART Universal Asynchronous Receiver/Transmitter |
| I2C two-wire serial bus                              | WDT Watchdog Timer                               |
| IAP Application Programming                          | WWDT Serial Watchdog Timer                       |
| IRQ Interrupt Request                                | XTAL External crystal                            |
| Latch-up                                             | VCO Voltage Controlled Oscillator                |
| LDO Low Dropout Regulator                            |                                                  |
| LED Light Emitting Diode                             |                                                  |
| LVR Low Voltage Reset                                |                                                  |
| MAC Media Access Control Layer                       |                                                  |



Revision History

| Version Date | Content              | Initial                                                |
|--------------|----------------------|--------------------------------------------------------|
| 1.0          | version released     | Jun.2024                                               |
| 1.1          | Updated to QFN48     | Jan.2025 Updated                                       |
| 1.2          | reference schematic, | updated operating frequency band and power consumption |

This version is an internal version and is for reference only.

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